Baseball Gear Customer Analysis & Product Association Report

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I. Abstract

This project analyzes order data from a baseball equipment retailer (2013–2016) using machine learning techniques: **RFM customer segmentation (K-Means clustering)** and **association rule mining (Apriori)**. It identifies two distinct customer types and seven strong product associations.

Key Findings:

1. **High-value customers account for only 0.19%** of the total, while **dormant customers make up 99.81%**. However, given the low-repurchase nature of durable sports goods, this pattern actually reflects a **loyalty-based e-commerce model** — there's no urgent need to aggressively acquire new customers.
2. Top product combinations (e.g., *541-baseball glove → 530-baseball glove*; *540-baseball glove → 529-baseball glove*) show a **lift of 16.05**, strongly suggesting bundling opportunities.

Business Recommendations:
Based on the analysis, we propose **five categories of actionable strategies** (eight specific actions):

1. **High-Value Customer Retention:** Offer bulk purchase discounts and VIP services to strengthen loyalty.
2. **Dormant Customer Reactivation:** Use email campaigns, content marketing, and after-sales events to re-engage.
3. **New Bundles:** Create two new product bundles guided by association rules.
4. **Algorithmic Cross-Selling:** Display real-time recommendations (e.g., "frequently bought together") during browsing.

5. **Online Shelf Optimization:** Place frequently co-purchased items (e.g., bat and bat bag) next to each other on the page.

II. Background & Objectives

2.1 Business Context

- **Company & Business:** We are a newly launched e-commerce store selling baseball equipment – primarily balls, bats & bat bags, uniforms, and other gear.
- **Data Scope:** Transaction records from July 2013 to July 2016, totaling over 60,000 order lines.
- **Growth Milestone:** A massive sales surge occurred in Q3 2015 (155× year-on-year growth), marking the beginning of our main sales period.
- **Pain Points:**
  - **Low repurchase rate** – we need to understand why and find ways to extend customer lifetime value.
  - **Wide product assortment** – we need to optimize cross-selling and product placement.

2.2 Project Goals

1. **Segment customers by value** to design differentiated marketing strategies.
2. **Uncover product association rules** to guide bundling and shelf layout.
3. **Deliver quantifiable, actionable business recommendations.**

III. Data Overview

3.1 Shopping Basket Construction

The raw data (order.csv) is an order-line table with daily granularity but **no explicit basket information.** To extract more

insights, we artificially constructed baskets using the following assumption:

All orders placed by the same customer on the same day are treated as a single shopping basket.

The resulting basket-size distribution is shown below.

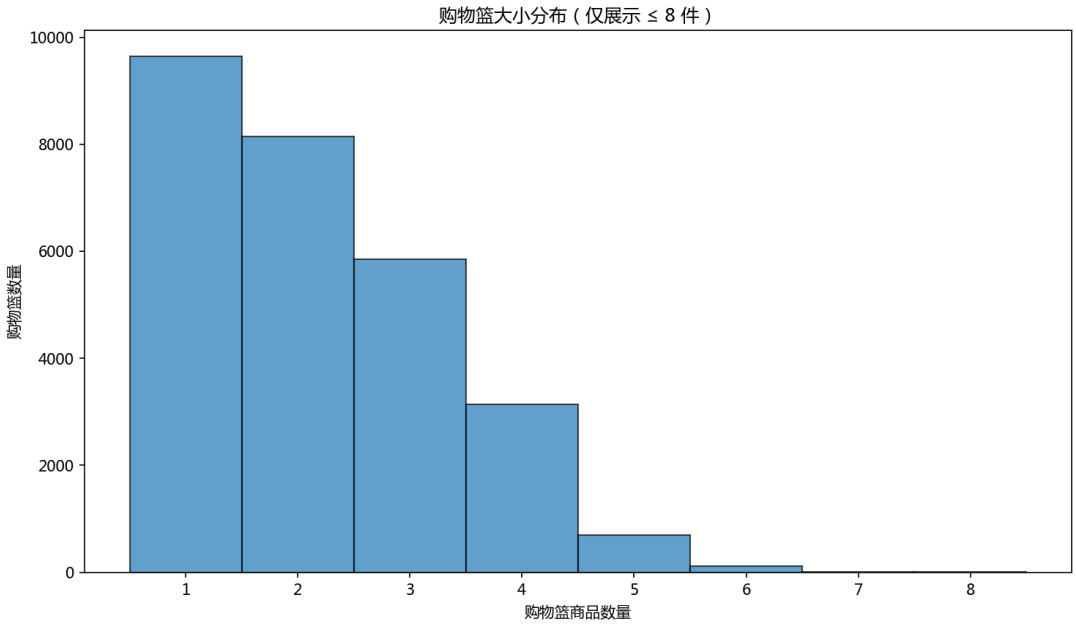

Figure 1 Basket size distribution

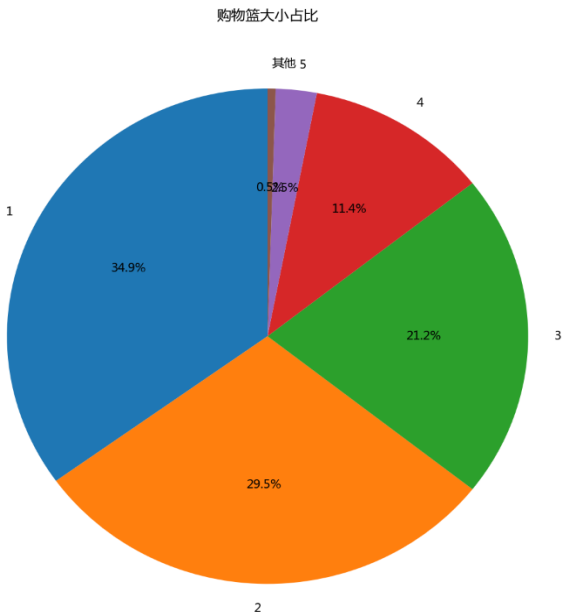

Figure 2 Basket size proportion

Observation: 34.9% of baskets contain only **one item**, while 65.1% contain **multiple items**. This reflects the nature of baseball equipment – balls, bats, gloves, etc., are highly complementary. Customers tend to buy a "set" to start playing, so multi-item purchases are the norm.

3.2 Key Business Metrics

Table 1: Key Business Metrics

Metric	Value
Total Orders	45,832
Total Customers	18,764
Total Product Types	159
Average Order Value	¥1,245
Overall Repurchase Rate	32%
Average Basket Size	2.186907
Average Revenue per Customer	¥2,096.34

Key Insights

1. Product mix is accessory-heavy, yet average order value is high.

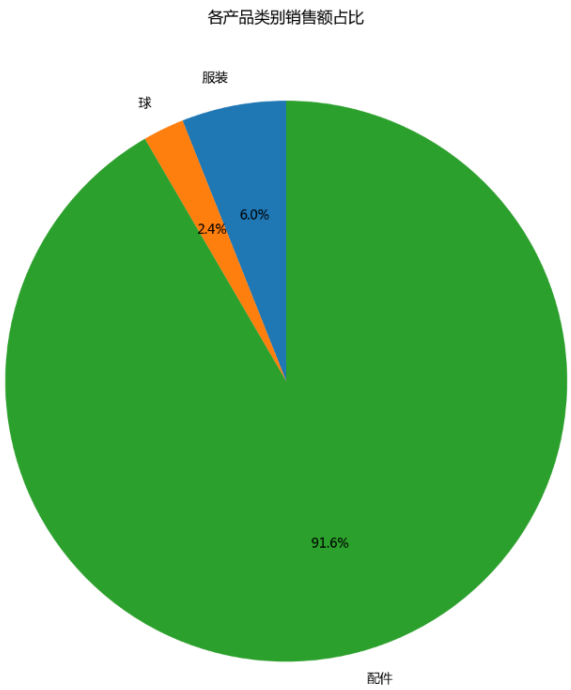

Figure 3 Sales share by product type

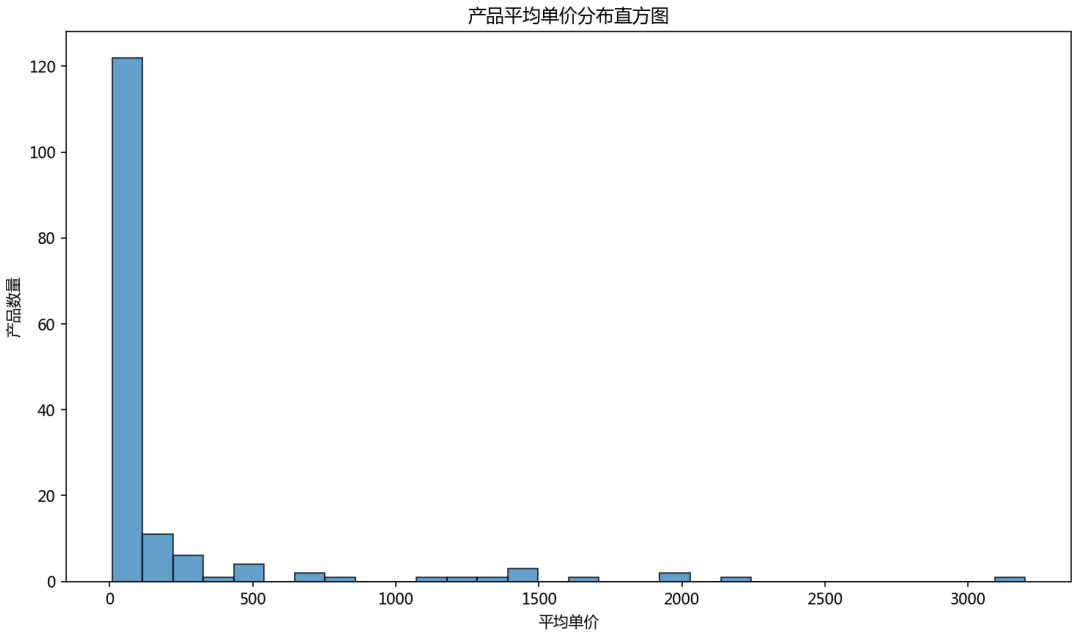

Figure 4 Average price distribution histogram

- **“Multi-user” nature:** A team may need only a couple of balls, but many bats, gloves, and bases – perfect for bundling.
- **Wide price range:** Professional bats can easily exceed ¥1,000, while basic gloves cost only tens of yuan. The

histogram (Fig. 4) shows that although most products are cheap (<¥500), a few high-end items pull the average order value up to ¥1,245. **High-end products, despite low volume, contribute high marginal profit.**

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2. **Low overall repurchase rate is not an urgent issue - it matches the product characteristics.**

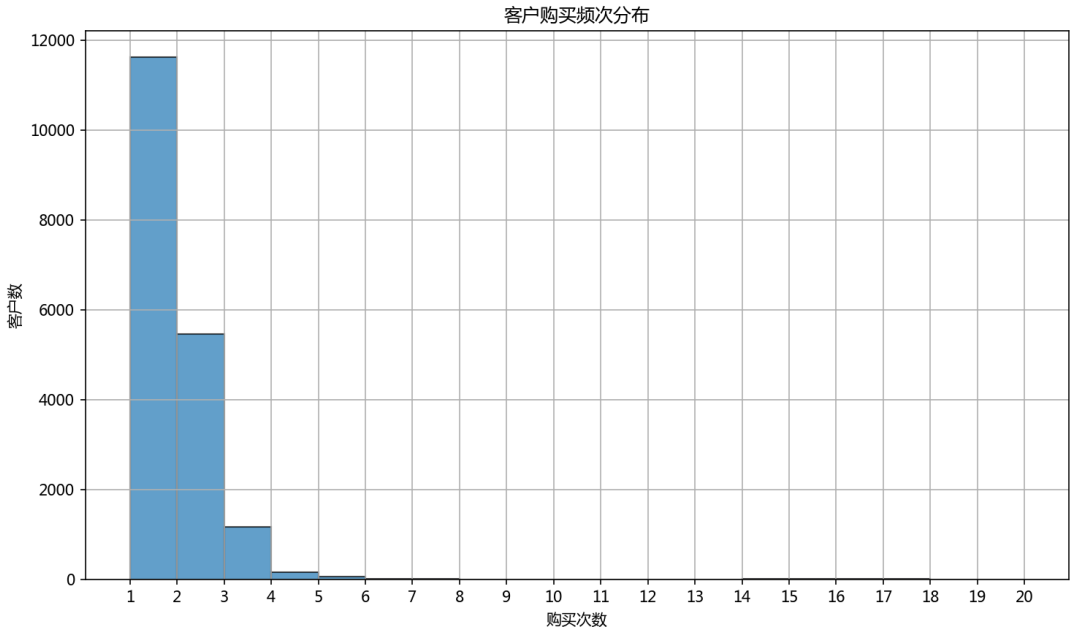

Figure 5 Customer purchase frequency distributio

- **Durable goods** naturally have long repurchase cycles.
- Our **basket construction assumption** (one basket per day) may also undercount repurchases if customers actually split orders across multiple transactions on the same day.

3.3 Sales Trend Over Time

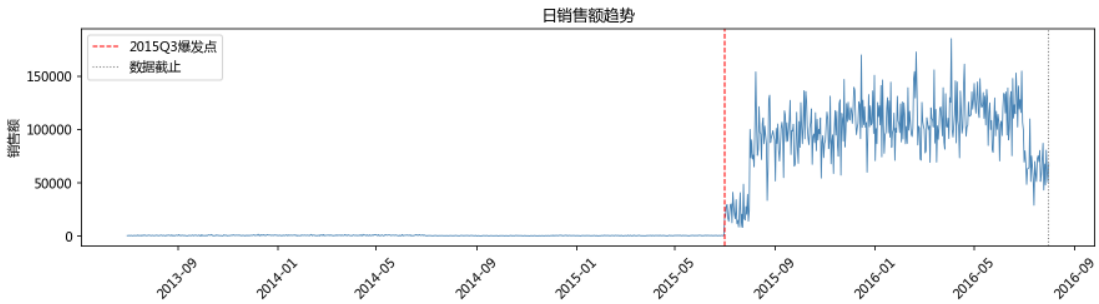

Figure 6 : Daily sales trend

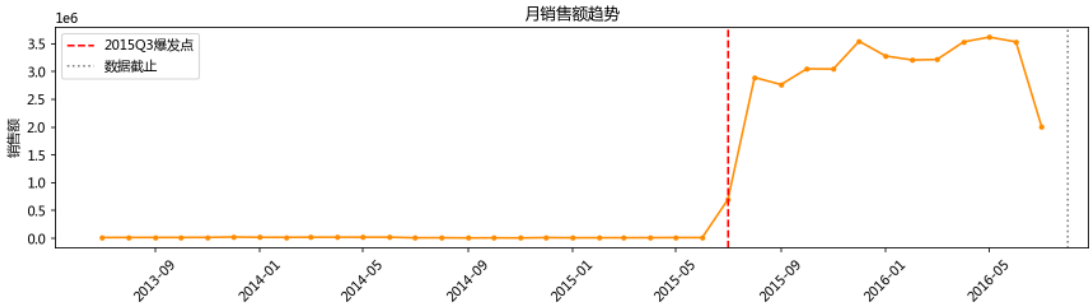

Figure 7 Monthly sales trend

- **Before Q3 2015:** Very low sales (business cold-start phase).
- **After Q3 2015:** Explosive growth – the vast majority of orders occurred after this point.

Implications:

- The data is highly skewed toward the recent period, so our analysis mainly reflects the **current business phase**.
- This recency ensures that insights remain relevant to today's operations.

IV. Customer Segmentation (RFM + K-Means)

4.1 Methodology

- **RFM Extraction:** For each customer we computed:
  - **R** (Recency) – days since last purchase
  - **F** (Frequency) – total number of purchases
  - **M** (Monetary) – total spending
    RFM is a widely used framework for customer value analysis; the three dimensions can be tailored to specific business contexts.
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- **K-Means Clustering:**
  After standardizing the RFM values, we applied K-Means. The **elbow method** and **silhouette score** both pointed to **K = 2** as the

optimal number of clusters.
K-Means is an unsupervised algorithm that partitions data into K clusters, each represented by its centroid, minimizing within-cluster distances – a classic approach for market segmentation.

4.2 Clustering Results

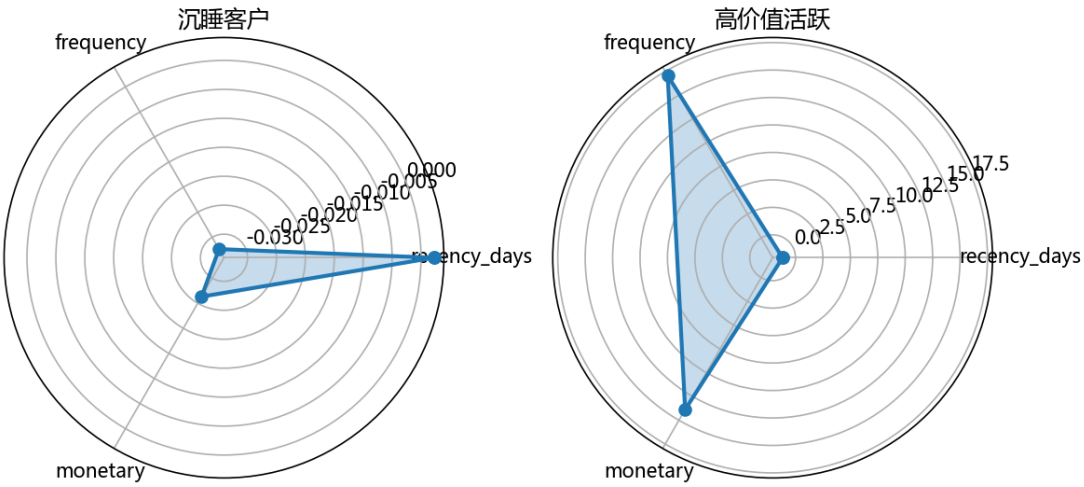

Figure 8 Radar chart of RFM features by cluster

Table 2: Cluster Characteristics

Cluster	Size (%)	Avg Recency (days)	Avg Frequency	Avg Monetary (¥)
Dormant Customers	99.81%	190.66	1.46	2023.02
High-Value Active	0.19%	18.51	19.77	40743.81

4.3 Business Interpretation

We defined high-value customers by the classic "two lows, one high" pattern: **low recency, high frequency, high monetary**. The clusters

clearly satisfy this, but with a striking imbalance – the vast majority are dormant.

Why are most customers dormant?
They are **individual enthusiasts**. Once they buy a complete set of gear, they have no immediate need for new products – dormancy here is perfectly natural.

Why do the 0.19% high-value customers exist?
Their extremely high frequency (≈20 purchases) and very low recency (≈19 days) suggest they are **not ordinary consumers**. They are likely **team purchasers** (coaches, team managers) or **family buyers** (parents buying for multiple children). Their "repurchases" are actually **bulk purchases for different people** at different times.

How to tailor strategies for the two groups?

Table 3: Marketing Strategies for Dormant Customers

Strategy	Example Actions	Goal
Content Marketing	Share baseball tips, gear maintenance guides, game schedules, etc., via social media / email.	Keep the brand top-of-mind so customers think of us when a need arises.
Upgrade / Replacement Reminders	Push notifications about new arrivals or seasonal offers; run trade-in events.	Proactively awaken demand before the old gear wears out and the customer turns to a competitor.

Table 4: Marketing Strategies for High-Value Active Customers

Strategy	Example Actions	Goal
Small-B Customer Care	Offer team discounts, bulk ordering fast-track, annual purchase contracts.	Secure recurring organizational purchases and leverage word-of-mouth to turn team members into individual customers.

V. Association Rules & Product Network

5.1 Methodology

- **Basket Matrix:** Created a binary matrix where each row is a basket and each column a product (presence = 1).
- **Apriori Algorithm:** Mined frequent itemsets with **minimum support 0.02** and **minimum confidence 0.5**.
- **Association Rules:** Retained rules with **lift > 2** and **confidence > 0.6**.
  Association rule mining uncovers relationships like "customers who buy X also buy Y" – classic for basket analysis.
- **Product Co-occurrence Network:** Used the **Louvain algorithm** to detect product communities.
  In this network, nodes are products and edges represent how often they appear together in baskets. Community detection reveals **natural product groups** – items frequently bought as a set.

5.2 Strongest Rules

Table 5: Top 7 Rules by Lift

Antecedent	Consequent	Support	Confidence	Lift
541-Baseball Glove	530-Baseball Glove	0.029256	0.543376	16.050221
540-Baseball Glove	529-Baseball Glove	0.029256	0.864171	16.050221
539-Baseball Glove	529-Baseball Glove	0.021544	0.695093	8.086391
537-Baseball Glove	528-Baseball Glove	0.021942	0.654428	7.613304
536-Bat & Bag	528-Bat & Bag	0.020132	0.654428	6.195635
479-Bat & Bag	477-Bat & Bag	0.055072	0.888434	5.782886
478-Bat & Bag	477-Bat & Bag	0.0612643	0.835555	5.438692

5.3 Product Network Analysis

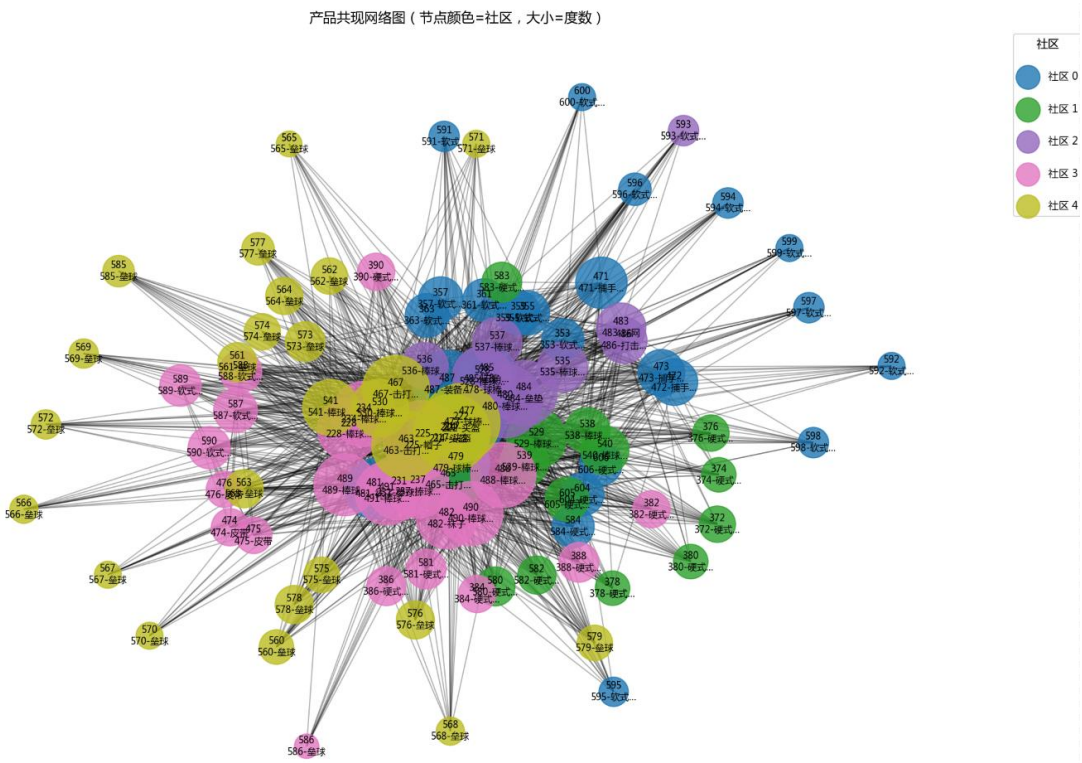

Figure 9 Product co-occurrence network graph

Table 6: Product Community Descriptions

Community	Theme	Product Types Included	Community Characteristics
Community 0 ●	General Ball Assortment	Dominated by balls (20 items: soft, hard, softball); few accessories (4: bat bags, equipment bags); no apparel.	Mostly various balls, plus carrying/storage items. Typical for teams/institutions buying different balls (practice, game, softball) in bulk along with storage gear.
Community 1 ●	Softball Personal Gear	Balls (8 – all soft); accessories (7 – gloves, helmets, net stands); apparel (5 –	Covers personal protection, hitting tools, and training aids – a complete set for a softball player, suitable for training or amateur

Community	Theme	Product Types Included	Community Characteristics
		batting gloves, catcher's gear, socks).	games.
Community 2 🟣	Team Uniforms & Balls	Apparel (13 – jerseys, caps, belts, socks); balls (13 – mostly hard, some soft, softball); no accessories.	Balanced mix of uniforms and balls, but lacks personal protective gear. Often purchased by teams for uniforming and game balls, while individuals bring their own gloves/helmets.
Community 3 🟣	Softball Equipment	Balls (13 – all softball); accessories (3 – gloves, helmets); apparel (2 – batting gloves).	Softball-centric, with basic personal gear – aimed at softball players.
Community 4 🟡	Hardball Training Gear	Accessories (8 – gloves, hitting tees, nets, bases); balls (6 – all hardball); no apparel.	Hardball + extensive training equipment (tees, nets for batting practice, bases for infield drills). Common for training facilities, schools, or clubs.

VI. Integrated Business Recommendations

6.1 Customer-Segmentation-Based Strategies

(A) High-Value Active Customers (0.19%)

1. Bulk Purchase Services:
   Offer team discounts, fast-track bulk ordering, and annual contracts. This streamlines their procurement and locks in steady revenue.
2. VIP Treatment:
   Send monthly new-arrival emails with exclusive discounts; build

a VIP community. With so few customers, the cost is negligible, but the payoff in loyalty and word-of-mouth is substantial.

(B) Dormant Customers (99.81%)

1. **Email Reactivation:**
   Send quarterly reactivation emails featuring high-confidence product pairs (e.g., glove + ball) with a 10–15% discount – a standard, low-effort reminder.
2. **Content Marketing:**
   Regularly publish baseball tips, gear care guides, game schedules, etc., on social media / official accounts to keep the brand engaged.
3. **After-Sales Events:**
   Run trade-in programs or "free trial of new gear with past order" events to stimulate replacement demand.

6.2 Association-Rule-Based Strategies

(A) Bundled Packages

Based on the discovered rules, we propose two bundles (product names interpreted from IDs, as detailed specs are unavailable):

- **Bundle A: "Master Your Game"**
  Contents: 541-Baseball Glove + 530-Baseball Glove
  Selling point: Two gloves for different scenarios – one durable for intense practice, one smooth for peak game performance.
- **Bundle B: "Sheathed Power"**
  Contents: 536-Bat + 528-Bat Bag
  Selling point: A great bat deserves a great bag – protect your weapon in style.

(B) Algorithmic Cross-Selling

On product detail pages, display "Frequently bought together" and "Customers who bought this also bought…" widgets, dynamically populated from the strong rules.

(C) Shelf Optimization

Place items from the same community (e.g., all products in Community 4) near each other on the website to increase serendipitous discovery.